Metropolitan University, Sylhet

Department of Computer Science Engineering

Mid Term Examination, Summer – 2023

Program: B.Sc. in CSE; Batch: 58

Course: CSE 123: Basic Electrical Engineering

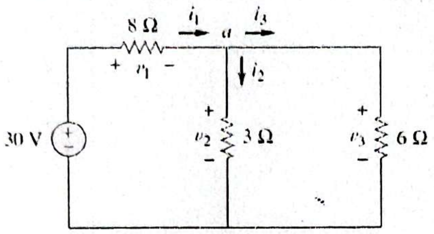
Time: 1.30 Hours **Marks: 30**

Answer any **three** questions out of four. All parts of each question must be answered sequentially.
Figures in the right margin indicate full marks.

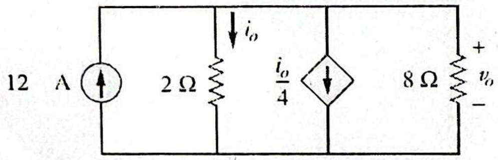
---

1. a) Explain Ohm's law with appropriate equation. 3

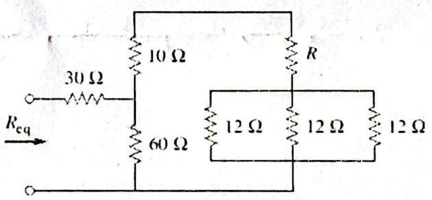
   b) Find i_1, i_2 and i_3 and v_1, v_2 and v_3 from the network below using KCL and KVL. 7



2. a) Find i_0 and v_0 from the following circuit. 5

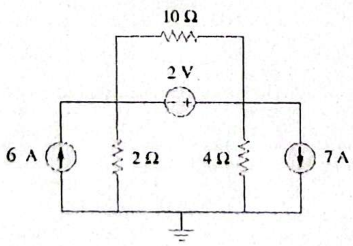


   b) For the network below R_{eq}=60 so find the value of R. 5

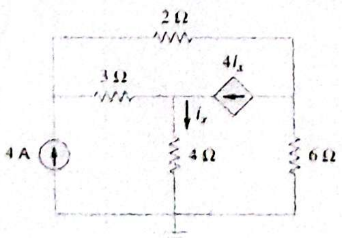


3. a) Explain Kirchhoff's laws with appropriate example. 4

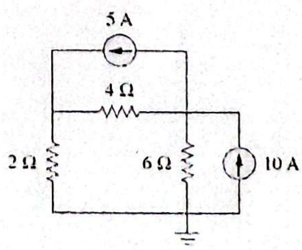
   b) Determine the node voltages for the following circuit 6



4. a) Find the node voltages for the circuits below. 10



1)



2)

Metropolitan University

Department of Computer Science and Engineering

Midterm Examination – Summer 2023

Program: **CSE:** Batch: 58(A-I)

Course: **GED 119**; Engineering Ethics and Cyber Law

Time: 1.5 Hour | Marks: 30

[Answer any three of the following questions]

1.	(a)	What is your understanding about cyber crime.	**2**
	(b)	Mention the types and categories of cyber crime with examples.	**3**
	(c)	Point out some common occurrence of cyber crime in Bangladesh	**4**
2.	**(a)**	What is cyber terrorism?	**3**
	(b)	Provide your relevant suggestion to reduce cyber crime in Bangladesh	**7**
3.		Discuss about different types of cyber crime against society?	**10**
4.	**(a)**	What is cyber jurisprudence?	**3**
	(b)	Describe the establishment and jurisdiction of the cyber tribunal and cyber appellate tribunal in Bangladesh.	**7**
5.	**(a)**	What are the objective, purpose and advantages of the Information and Communication Technology Act, 2006 (Act No. 39 of 2006)	**7**
	(b)	Write down the short note on: • Digital signature • Data	**1.5x2 =3**

METROPOLITAN UNIVERSITY
Department of Computer Science and Engineering
Midterm Examination, Summer 2023
Programme: CSE, Batch: 58th
Course Code: Eng 115, Course Title: English Language II

Time: 1 hour 30 minutes — Marks: 30

Read the following passage and answer questions 1 and 2:

I contain multitudes

Wendy Moore reviews Ed Yong's book about microbes

Microbes, most of them bacteria, have populated this planet since long before animal life developed and they will outlive us. Invisible to the naked eye, they are ubiquitous. They inhabit the soil, air, rocks and water and are present within every form of life, from seaweed and coral to dogs and humans. And, as Yong explains in his utterly absorbing and hugely important book we mess with them at our peril.

Every species has its own colony of microbes, called a 'microbiome', and these microbes vary not only between species but also between individuals and within different parts of each individual. What is amazing is that while the number of human cells in the average person is about 30 trillion, the number of microbial ones is higher – about 39 trillion. At best, Yong informs us, we are only 50 per cent human. Indeed, some scientists even suggest we should think of each species and its microbes as a single unit, dubbed a 'holobiont'.

In each human there are microbes that live only in the stomach, the mouth or the armpit and by and large they do so peacefully. So 'bad' microbes are just microbes out of context. Microbes that sit contentedly in the human gut (where there are more microbes than there are stars in the galaxy) can become deadly if they find their way into the bloodstream. These communities are constantly changing too. Every time we eat, we swallow a million microbes in each gram of food; we are continually swapping microbes with other humans, pets and the world at large.

For most of human history we had no idea that microbes existed. The first man to see these extraordinarily potent creatures was a Dutch lens-maker called Antony van Leeuwenhoek in the 1670s. Using microscopes of his own design that could magnify up to 270 times, he examined a drop of water from a nearby lake and found it teeming with tiny creatures he called 'animalcules'. It wasn't until nearly two hundred years later that the research of French biologist Louis Pasteur indicated that some microbes caused disease. It was Pasteur's 'germ theory' that gave bacteria the poor image that endures today.

New research is now unravelling the ways in which bacteria aid digestion, regulate our immune systems, eliminate toxins, produce vitamins, affect our behaviour and even combat obesity. But we are facing a growing problem. Our obsession with hygiene, our overuse of antibiotics and our unhealthy, low-fibre diets are disrupting the bacterial balance and may be responsible for soaring rates of allergies and immune problems, such as inflammatory bowel disease (IBD).

The most recent research actually turns accepted norms upside down. For example, there are studies indicating that the excessive use of household detergents and antibacterial products actually destroys the microbes that normally keep the more dangerous germs at bay. Other studies show that keeping a dog as a pet gives children early exposure to a diverse range of bacteria, which may help protect them against allergies later. (Abbreviated)

1. Answer the following questions: **1×5=5**

a) Why does Yong say we are only 50% human?

b) When can the microbes in the gut become deadly?
c) What did Antony van Leeuwenhoek find in the droplet of water that he collected from a nearby lake?
d) What does the writer say gave microbes a bad image?
e) How does keeping pets protect children from allergies?

2. Paraphrase the following extract from the passage: 06

The most recent research actually turns accepted norms upside down. For example, there are studies indicating that the excessive use of household detergents and antibacterial products actually destroys the microbes that normally keep the more dangerous germs at bay. Other studies show that keeping a dog as a pet gives children early exposure to a diverse range of bacteria, which may help protect them against allergies later.

3. Correct the following sentences: 0.5×8=4

a) I'm not sure,but I might leave the wallet in your car yesterday.
b) You didn't need to prepare this food, because this food will be wasted now.
c) You could tell your mother but you didn't.
d) I can hear footsteps, so someone must be comein this direction.
e) She must not be Shaila's sister; they look totally different.
f) The siblings or their father are responsible for this incident.
g) 20 miles are a long distance for commuting.
h) I would be given him the money, but I know he will never pay it back.

4. Fill in the gaps with phrases from the box. 0.5×10=5

make sense of, pick things up, brush up on, rack my brains, on the tip of my tongue, keep up with, stuck at, keep it up, slipped my mind, keep your mind on,

As I had planned a holiday in Spain, I decided to (a) ______ my Spanish before going there, so I enrolled on a Spanish language course. At first I had to (b) ______ to remember things, and I couldn't utter the words which were (c) ______. And the words I learnt often (d) ______. I couldn't (e) ______ the grammar, either. So it was beginning to be hard for me to (f) ______ the other students. When you work hard all day, it's not easy to (g) ______ a difficult subject in the evening. But I (h) ______ it and began to (i) ______ more quickly. All I have to do now is (j) ______.

5. Briefly discuss any ONE of the following topics. Write at least 200 words. 10

a) Pros and cons of having a part-time job during student life.

b) Importance of time management.

Metropolitan University

Department of Computer Science and Engineering (CSE)

Midterm Examination – Summer 2023

Program: CSE; Batch: 58

Course: MAT 123; Differential Equations and Laplace Transforms.

Time: 1 Hour 30 Minutes Marks: 30

[Answer any three of the following questions]

1. Consider the differential equation

$$(y^2 + 2xy)dx - x^2 dy = 0$$

(a) Define Exact Differential Equation and show that this equation is not exact. 3

(b) Multiply the given equation through by y^n, where n is an integer, and then determine n so that y^n is an integrating factor of the given equation 3

(c) Multiply the given equation through by the integrating factor found in (b) and solve the resulting exact equation. 4

2. (a) Define Separable Equations and Homogeneous Equations. 2

(b) Solve the following differential equations. 8

(i) $2r(s^2+1)dr + (r^4+1)ds = 0.$

(ii) $(y+2)dx + y(x+4)dy = 0,\ y(-3) = -1.$

3. (a) Define Linear Equations and Bernoulli's Equations. 2

(b) Solve the following differential equations. 8

(i) $x\frac{dy}{dx} + \frac{2x+1}{x+1}y = x - 1.$

(ii) $\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}$

4. (a) Define Initial Value Problem and given that every solution of 6

$$x^3\frac{d^3y}{dx^3} - 3x^2\frac{d^2y}{dx^2} + 6x\frac{dy}{dx} - 6y = 0$$

may be written in the form $y = c_1x + c_2x^2 + c_3x^3$ for some choice of the arbitrary constants c_1, c_2, and c_3, solve the initial-value problem consisting of the above differential equation plus the three conditions

$$y(2) = 0,\ y'(2) = 2,\ y''(2) = 6.$$

(b) Find the differential equation of the family of curves $y = e^x(A\cos x + B\sin x)$, where A and B are arbitrary constants. 4

Metropolitan University

Department of Computer Science and Engineering

Midterm Examination – Summer 2023

Program: **CSE**: Batch: **58**

Course: **PHY 123**: **Physics II**

Time: 1.5 Hour **Marks:** 30

[Answer any **three** of the following questions]

1. (a) Can we have central minima? Explain. **3**

   (b) Why is monochromatic light used in the double slit experiment? **2**

   (c) What is the wavelength of light falling on double slits separated by 2 μm if the third-order maximum is at an angle of 60°? **5**

2. (a) Young's double-slit experiment breaks a single light beam into two sources. Would the same pattern be obtained for two independent sources of light, such as the headlights of a distant car? Explain. **3**

   (b) In the double slit experiment what would happen if white light were used? **2**

   (c) What is the smallest separation between two slits that will produce a second-order maximum for any visible light? **5**

3. (a) You are conducting a single-slit diffraction experiment with light of wavelength. What appears, on a distant viewing screen, at a point at which the top and bottom rays through the slit have a path length difference equal to **3**

   - 2λ and
   - 1.5λ?

   (b) Can you determine the wavelength of an X-ray using diffraction grating? Explain. **2**

   (c) A grating has 1400 lines/mm. How many orders of the entire visible spectrum (380–750 nm) can it produce in a diffraction experiment, in addition to the central maxima? **5**

4. (a) Explain diffraction using Huygens principle. **4**

   (b) Red light of wavelength of 700 nm falls on a double slit separated by 400 nm. **6**

      1. At what angle is the first-order maximum in the diffraction pattern?
      2. What is unreasonable about this result?
      3. Which assumptions are unreasonable or inconsistent?

7.1428×10^{-7}

Metropolitan University, Sylhet

Department of Computer Science and Engineering
Summer Midterm Examination – 2023
Program: B.Sc. in CSE; CSE(DAY)
Batch: 58
Course: CSE 121: Structured Programming

Time: 1.5 Hours **Marks: 30**

Answer any **three** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

---

1. a) Provide three rules for identifiers in C programming. 3

b) Let we have an integer variable i. Your system needs to increment i's value a little bit faster. Which one you will use ++i or i++? Describe why. 3

c) Given the following variables: 4

```
int i = 3, j = 5;
float m = 8.01;
char c = 'c', d = 'd';
```

Identify whether the below conditions are true or false.

i) (3 * i − 2 * j) != 0
ii) 10 * (i + j) > d
iii) (2 * d − c) % 2 == i
iv) m > j + i

2. a) Find the output of the following program. 5

```
#include <stdio.h>
int main()
{
    int x = 10, y = 5;
11 - ++x;                 r12.
    printf("x = %d\n",x++);
    x += y; - 17                      r4
    printf("x = %d, y = %d\n", x, y--);
    y %= ++x;
    printf("x = %d, y = %d\n", x, y);
    return 0;
}
```

b) Find the error of the following program and write the correct program. 5

```
#include <stdio.h>
int main()
{
   int x = 10, int y = 20;
   x = x + y;
   y = x - y,
   x = x - y;
   printf("x: %d, y: %d\n, x, y");
   return 10;
}
```

3. a) What are the primary classifications of data types in c? Is "string" the data type of c language? 3

b) Write a C program to solve the following problem. 7
Given an array of 10 integers, write a program to determine the largest element in the array.

Sample Input:
10 46 50 27 -30 76 20 12 -60 38

Sample Output:
76

4. a) Write the output of the below code: 4

```
#include<stdio.h>
int main()
{
   int first=0,second=1;
   while(second < 5)
   {
      printf("%d\n",second);
      first = second;
      second += first;
   }
}
```

b) Write a C program which determines whether a character is vowel. If vowel, print "Yes" otherwise "No". 6